

MD ABDUR RABBI

Mechatronics Engineer | Robotics · Automation · Embedded Systems

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CAREER OBJECTIVE

Mechatronics Engineering graduate of KUET with strong analytical and research skills, built through a comparative reinforcement-learning thesis and hands-on with PLC, embedded C, STM32/ESP32, computer vision-based projects. Aims to design and build automation and embedded systems for real industrial and manufacturing problems.

EDUCATION

Khulna University of Engineering & Technology (KUET) Mar 2022 – Aug 2026
B.Sc. in Mechatronics Engineering — CGPA: 3.57 / 4.00
Alamdanga Govt. College, Chuadanga — HSC (Science) — GPA: 5.00/5.00 2020
Alamdanga Academy, Chuadanga — SSC (Science) — GPA: 5.00/5.00, Govt. General Scholarship 2018

TECHNICAL SKILLS

Robotics & Simulation: NVIDIA Isaac Sim, Isaac Lab, computer vision, control systems, PID Control
Embedded & IoT: Arduino, ESP32, STM32 Nucleo, drivers, sensors & actuators, IoT dashboards
CAD & Simulation: SolidWorks, MATLAB, Simulink, Proteus
Programming: Python (NumPy, Matplotlib, OpenCV), C (Embedded C), Linux (basics)
Other: PLC (laboratory training), HMI, Git & GitHub, Microsoft Office (Word, Excel, PowerPoint)

PROJECTS

Undergraduate Thesis — Comparative Evaluation of Constrained and Unconstrained Deep Reinforcement Learning for Safe Robotic Grasping on a UR5e Manipulator in Simulation 2026

- Trained and compared PPO vs. PPO-Lagrangian policies for UR5e grasping in Isaac Lab.
- Cut safety-constraint violations by 80% with a 99% grasp success rate.
- Simulated my research in NVIDIA Isaac Sim/Isaac Lab on Linux.

Automated Paper Grading and Dispensing System — MTE 4108 Sessional, 5-member team project 2025

- Helped design the NEMA17 lead-screw paper-feed mechanism and debugged hardware.
- Integrated Raspberry Pi control for OCR-based grading (Google Cloud Vision).

Gesture-Controlled Car — computer-vision-based control (Academic Project) 2025

- Built a car driven entirely by hand gestures using OpenCV & computer vision.
- Python/OpenCV camera-based recognition translated hand movements into wireless drive commands.

IoT Real-Time Weather Monitoring Station — three independent platform builds 2024

- Built a real-time weather station (temperature, humidity, pressure, light) using different sensors and actuators.
- Implemented separately on Arduino Uno, ESP32, and STM32 with a local LCD and a Wi-Fi IoT web dashboard.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Core Organizer & Event Lead — CALIBRATION 2.0, National Robotics Festival, KUET 2026

- Led Logistics & Kit Management segments with more than 1200 participants and organisers.
- Co-led the RC Speed Battle team on track design and event management.
- Contributed to whole event planning and budgeting and also directed teams of volunteers.

Co-Lead, Drivetrain — Team Kilo Flight (Formula Student Team), KUET Jun 2024 – Sep 2025

- Worked on drivetrain and electrical powertrain design.
- Conducted workshop sessions for junior batches.
- Helped the team place in multiple online quiz rounds, including Germany, Hungary, and India.

Member, Sports Committee — Residential Hall Committee, KUET 2024 – 2025

- Organized 5 inter-batch tournaments of both indoor and outdoor games.
- Maintained sporting facilities.
- Served 3 times as the dining manager.

Hardware Section Co-Lead — KAAVIN (Aircraft Design Team), KUET Nov 2024 – Feb 2025

- Guided the hardware section for aircraft builds targeting international design competitions.

AWARDS & ACTIVITIES

- Got Technical Scholarship for academic excellence in all academic sessions.
- **Champion** — Inter-Batch Badminton (Men's Singles), Dept. of Mechatronics Engineering.
- Inter-Department Tournaments 2026: Player (Badminton), Team Coach (Futsal), Player (Cricket).

Languages: Bangla (Native) · English (Fluent — speaking & writing)